

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 5

- (1) Suppose that $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are functions such that $g \circ f = \text{id}_X$. Show that f is injective (that is, one-to-one) and that g is surjective (that is, onto). Show by example that neither f nor g need be bijective.

Solution. We are told $g(f(x)) = x$ for all $x \in X$.

Injectivity of f : If $f(x_1) = f(x_2)$, then $x_1 = g(f(x_1)) = g(f(x_2)) = x_2$;

Surjectivity of g : Every $x \in X$ can be written as $g(f(x))$, and hence is an output of g .

- (2) Suppose that $T : V \rightarrow W$ is an invertible F -linear transformation of F -vector spaces. Show that the inverse function $T^{-1} : W \rightarrow V$ is also a linear transformation.

Solution. We have to check:

- (Preservation of sums) $T^{-1}(w_1 + w_2) = T^{-1}(w_1) + T^{-1}(w_2)$

This can be checked after applying T to both sides (why?) and using the fact that T preserves sums:

$$\begin{aligned} T(T^{-1}(w_1 + w_2)) &= w_1 + w_2 \\ T(T^{-1}(w_1) + T^{-1}(w_2)) &= T(T^{-1}(w_1)) + T(T^{-1}(w_2)) = w_1 + w_2. \end{aligned}$$

The right hand sides here are clearly equal.

- (Preservation of scalar products) $T^{-1}(a \cdot w) = a \cdot T^{-1}(w)$ for $a \in F, w \in W$.

Once again, we check this after applying T and using the fact that T preserves scalar products:

$$\begin{aligned} T(T^{-1}(a \cdot w)) &= a \cdot w \\ T(a \cdot T^{-1}(w)) &= a \cdot T(T^{-1}(w)) = a \cdot w. \end{aligned}$$

We will prove the following result in class:

Proposition 1 (Rank-Nullity). If $T : V \rightarrow W$ is a linear transformation of F -vector spaces, then

$$\dim \ker(T) + \dim \text{im}(T) = \dim W.$$

- (3) Decide if the following can exist. Justify your response by explaining why it cannot or by giving an example when it does:

- (a) A linear transformation $T : F^4 \rightarrow F^4$ with $\ker(T) = \text{im}(T)$.

Solution. Since $\dim \ker(T) + \dim \text{im}(T) = 4$, this means that $\ker(T)$ and $\text{im}(T)$ must both be two-dimensional. An example is given by the linear transformation associated with the matrix

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Both the image and kernel of this linear transformation are spanned by the standard basis vectors e_1, e_2 .

- (b) A linear transformation $U : F^5 \rightarrow F^5$ with $\ker(T) = \text{im}(T)$.

Solution. Since 5 is not even, this is not possible.

- (4) Suppose that $S, T : F^n \rightarrow F$ are linear transformations with the same kernel. Show that there exists $c \in F^\times = F \setminus \{0\}$ such that $cS = T$.

Solution. Both S and T correspond to $1 \times n$ -matrices

$$B = [S(e_1) \quad \cdots \quad S(e_n)] ; A = [T(e_1) \quad \cdots \quad T(e_n)]$$

where $S(e_i) = s_i \in F$ and $T(e_i) = t_i \in F$. In terms of these matrices, we have

$$S \left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \right) = s_1 a_1 + \cdots + s_n a_n ; T \left(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \right) = t_1 a_1 + \cdots + t_n a_n$$

So such a vector is in $\ker(S)$ precisely when $s_1 a_1 + \cdots + s_n a_n = 0$ and is in $\ker(T)$ when $t_1 a_1 + \cdots + t_n a_n = 0$.

By rank-nullity, we have $\dim \ker(T) + \dim \text{im}(T) = n$. Since $\text{im}(T)$ is a subspace of F , its dimension is at most 1.

There are now two possibilities:

- $\dim \text{im}(T) = 0$: In this case, T is the trivial or zero linear transformation, and $\ker(T) = F^n$. Since we are assuming that $\ker(S) = \ker(T)$, this tells us that S is also the zero linear transformation, and so $cS = T$ for all $c \in F$.
- $\dim \text{im}(T) = 1$: In this case, $\dim \ker(T) = n - 1 = \dim \ker(S)$. Choose a basis $(v_1, \dots, v_{n-1})$ for $\ker(T)$. We can extend this to a basis $(v_1, \dots, v_n)$ for F^n (why?). We must have $t_n = T(v_n) \neq 0$ (why?). Moreover, any element of F^n can be written uniquely in the form

$$v = a_1 v_1 + \cdots + a_n v_n \Rightarrow T(v) = T(a_1 v_1 + \cdots + a_n v_n) = \sum_i a_i T(v_i) = a_n t_n,$$

because $T(v_i) = 0$ for $i = 1, \dots, n-1$ (why?).

Now, since $\ker(S) = \ker(T)$, we have

$$S(v) = \sum_i a_i S(v_i) = a_n S(v_n).$$

If $s_n = S(v_n)$, then $s_n \neq 0$ (why?) If we set $c = s_n^{-1} t_n$, then we see

$$(cS)(v) = c \cdots S(v) = s_n^{-1} t_n (s_n a_n) = t_n a_n = T(v).$$

- (5) Decide if the following are true for a linear transformation $T : V \rightarrow W$:

- (a) If T is injective and $v_1, \dots, v_n \in V$ are linearly independent, then $T(v_1), \dots, T(v_n) \in W$ are also linearly independent.

Solution. True. If we had a relation $a_1 T(v_1) + \cdots + a_n T(v_n) = 0$, then we would obtain

$$T(a_1 v_1 + \cdots + a_n v_n) = 0.$$

Since T is injective, this is only possible if $a_1 v_1 + \cdots + a_n v_n = 0$, and since the vectors are linearly independent, this implies that $a_1 = a_2 = \cdots = a_n = 0$.

- (b) If T is surjective and $v_1, \dots, v_n \in V$ are linearly independent, then $T(v_1), \dots, T(v_n) \in W$ are also linearly independent.

Solution. False. Take the linear transformation $T : F^3 \rightarrow 2$ given by $T(e_1) = e_1, T(e_2) = e_2$ and $T(e_3) = e_2$. This is surjective (why?) but $T(e_2), T(e_3)$ are not linearly independent, since they are equal.

- (c) If T is injective and $v_1, \dots, v_n \in V$ span V , then $T(v_1), \dots, T(v_n) \in W$ span W .

Solution. False. Take the linear transformation $T : F^2 \rightarrow F^3$ given by $T(e_1) = e_1$ and $T(e_2) = e_2$. Then e_1, e_2 span F^2 but their images under T cannot span F^3 .

- (d) If T is surjective and $v_1, \dots, v_n \in V$ span V , then $T(v_1), \dots, T(v_n) \in W$ span W .

Solution. True. Every element of W is of the form $T(v)$, and every $v \in V$ is a linear combination $v = a_1v_1 + \cdots + a_nv_n$. Applying T and using linearity shows that every element of W is a linear combination $a_1T(v_1) + \cdots + a_nT(v_n)$.

- (6) Suppose that $T : V \rightarrow W$ is a linear transformation of finite dimensional F -vector spaces. Show:

- (a) If T is injective, then $\dim V \leq \dim W$.

Solution. We say in part (a) of the above problem that T takes linearly independent sets to linearly independent sets. In other words, if $(v_1, \dots, v_n)$ is a basis for V , then $T(v_1), \dots, T(v_n)$ are linearly independent. This means that we must have $\dim V = n \leq \dim W$ (why?).

- (b) If T is surjective, then $\dim V \geq \dim W$.

Solution. Similarly, by part (d) of the previous problem, a surjective T takes spanning sets to spanning sets, and so, if $v_1, \dots, v_n$ is a basis for V , then $T(v_1), \dots, T(v_n)$ is a spanning set for W , and so we must have $n \geq \dim W$ (why?).

Definition 1. If $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$ are matrices corresponding to linear transformations $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^p \times \mathbb{R}^n$, then the **matrix product** $AB \in M_{m \times p}$ is the matrix corresponding to the composition $T \circ S : \mathbb{R}^p \rightarrow \mathbb{R}^m$.

- (7) If B is given as a column matrix $[v_1 \ \cdots \ v_p]$, show that AB is the column matrix $[Av_1 \ \cdots \ Av_p]$.

Solution. By definition of the matrix product Av in Lecture 16, we have $Av = T(v)$, where A corresponds to the linear transformation T . The *columns* of the matrix A are given by $T(e_1), \dots, T(e_n)$.

Similarly, $Bw = S(w)$, where B corresponds to S , and the columns of B are given by $v_1 = S(e_1), \dots, v_p = S(e_p)$. Now, AB is the matrix satisfying

$$(AB)w = (T \circ S)(w) = T(S(w)).$$

Its columns are now given by $T(S(e_1)), \dots, T(S(e_p))$. But from what we have just seen, we have $Av_i = T(v_i) = T(S(e_i))$ for $i = 1, \dots, p$.

Definition 2. A linear transformation $T : V \rightarrow W$ is an **isomorphism** if it is invertible. In this case, by problem 2, $T^{-1} : W \rightarrow V$ is also an isomorphism. We will emphasize the isomorphism property by writing $T : V \xrightarrow{\sim} W$.

- (8) Show that $U : F^n \rightarrow V$ is an isomorphism if and only if the vectors $U(e_1), \dots, U(e_n) \in V$ are a basis.

Solution. If U is an isomorphism, then it is both injective and surjective, and so parts (a) and (d) of problem 5 tell us that $(U(e_1), \dots, U(e_n))$ is a basis for V .

Conversely, if this is a basis, then we see:

- (a) $U(e_1), \dots, U(e_n)$ span V . This tells us that every element of V can be written in the form $v = a_1U(e_1) + \cdots + a_nU(e_n) = U(\sum_i a_ie_i)$. In other words, U is surjective.
- (b) $U(e_1), \dots, U(e_n)$ are linearly independent. This tells us that if

$$U(\sum_i a_ie_i) = a_1U(e_1) + \cdots + a_nU(e_n) = 0$$

then $a_1 = a_2 = \cdots = 0$. This is telling us exactly that U is injective. Combining the two now tells us that U is an isomorphism.

Definition 3. Suppose that $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis for V corresponding to an isomorphism $U : F^n \xrightarrow{\cong} V$ with $U(e_i) = v_i$. Given a linear transformation $T : V \rightarrow V$, the **matrix representation for T with respect to $\mathcal{B}$** , or simply **$\mathcal{B}$ -matrix**, is the matrix $[T]_{\mathcal{B}} \in M_{n \times n}(F)$ representing the composition

$$F^n \xrightarrow[\cong]{U} V \xrightarrow{T} V \xrightarrow[\cong]{U^{-1}} F^n$$

- (9) Let $D : \text{Poly}_{\leq 4}(\mathbb{R}) \rightarrow \text{Poly}_{\leq 4}(\mathbb{R})$ be the derivative function. Find $[D]_{\mathcal{B}}$ with respect to the following bases:

(a) $\mathcal{B} = \{1, x, x^2, x^3, x^4\}$.

Solution. Unwinding definitions, this amounts to writing down what happens under D to the basis elements. See Lecture 16. We compute

$$D(1) = 0$$

$$D(x) = 1 \cdot 1$$

$$D(x^2) = 2x = 0 \cdot 1 + 2 \cdot x$$

$$D(x^3) = 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2$$

$$D(x^4) = 4x^3 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 + 4 \cdot x^3$$

The matrix now looks like

$$[D]_{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(b) $\mathcal{B} = \{1, x-1, (x-1)^2, (x-1)^3, (x-1)^4\}$

Solution. It's the same matrix as for (a)!